

Series GEFH1/C



SET ~ 2



प्रश्न-पत्र कोड
Q.P. Code 55/C/2

रोल नं.

Roll No.



परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book. *

भौतिक विज्ञान (सैद्धान्तिक)

PHYSICS (Theory)

निर्धारित समय : 3 घण्टे

अधिकतम अंक : 70

Time allowed : 3 hours

Maximum Marks : 70

नोट / NOTE :

- (i) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 27 हैं।
Please check that this question paper contains 27 printed pages.
- (ii) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- (iii) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 35 प्रश्न हैं।
Please check that this question paper contains 35 questions.
- (iv) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
Please write down the serial number of the question in the answer-book before attempting it.
- (v) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.



General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) *This question paper contains **35** questions. **All** questions are **compulsory**.*
- (ii) *This question paper is divided into **five** Sections – **A, B, C, D** and **E**.*
- (iii) *In **Section A** – Questions no. **1** to **18** are Multiple Choice (MCQ) type questions, carrying **1** mark each.*
- (iv) *In **Section B** – Questions no. **19** to **25** are Very Short Answer (VSA) type questions, carrying **2** marks each.*
- (v) *In **Section C** – Questions no. **26** to **30** are Short Answer (SA) type questions, carrying **3** marks each.*
- (vi) *In **Section D** – Questions no. **31** to **33** are Long Answer (LA) type questions carrying **5** marks each.*
- (vii) *In **Section E** – Questions no. **34** and **35** are case-based questions carrying **4** marks each.*
- (viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 3 questions in Section D and 2 questions in Section E.*
- (ix) *Use of calculators is **not** allowed.*

Use the following values of physical constants, if required :

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\text{Mass of electron (} m_e \text{)} = 9.1 \times 10^{-31} \text{ kg}$$

$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

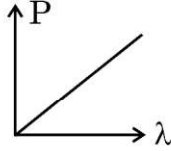
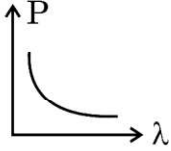
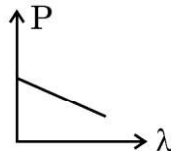
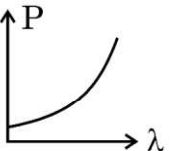
$$\text{Boltzmann constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}$$



SECTION A

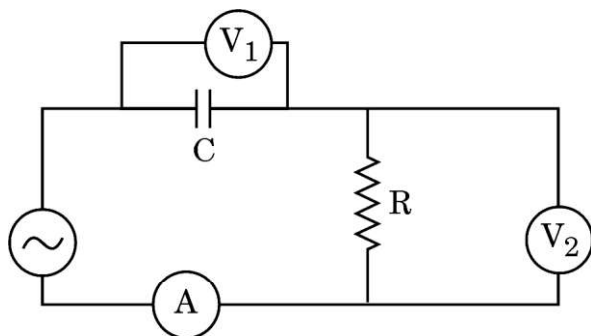
1. In a series LCR circuit, the capacitance is changed from C to $C/4$. For the resonant frequency to remain unchanged, the inductance should be changed from L to nL , where n is :
(a) $\frac{1}{2}$ (b) 2
(c) 4 (d) $\frac{1}{4}$
2. An electron with velocity $\vec{v} = (v_x \hat{i} + v_y \hat{j})$ moves through a magnetic field $\vec{B} = (B_x \hat{i} - B_y \hat{j})$. The force $\vec{F}$ on the electron is : (e is the magnitude of its charge)
(a) $-e (v_x B_y - v_y B_x) \hat{k}$ (b) $e (v_x B_y - v_y B_x) \hat{k}$
(c) $-e (v_x B_y + v_y B_x) \hat{k}$ (d) $e (v_x B_y + v_y B_x) \hat{k}$
3. A small bar, when placed near a magnet is repelled by it. This is because the bar is made of :
(a) Iron (b) Copper
(c) Aluminium (d) Nickel
4. When a negative charge ($-Q$) is brought near one face of a metal cube, the :
(a) cube becomes positively charged
(b) cube becomes negatively charged
(c) face near the charge becomes positively charged and the opposite face becomes negatively charged
(d) face near the charge becomes negatively charged and the opposite face becomes positively charged
5. A metal detector is based on :
(a) Self-induction (b) Mutual induction
(c) Electrical resonance (d) Power transmission



6. In Young's double-slit experiment, the intensity on the screen is I_0 at a point where path difference is λ . The intensity at the point where path difference is $\frac{\lambda}{4}$ is :
- (a) $\frac{I_0}{4}$ (b) $\frac{I_0}{2}$
(c) I_0 (d) zero
7. Which of the following figures represents the variation of a particle's momentum with the de Broglie wavelength associated with it ?
- (a)  (b) 
- (c)  (d) 
8. The binding energy per nucleon of ${}_8\text{O}^{16}$ is 7.97 MeV and that of ${}_8\text{O}^{17}$ is 7.75 MeV. The energy (in MeV) required to remove a neutron from ${}_8\text{O}^{17}$ is :
- (a) 0.42 MeV (b) 7.86 MeV
(c) 4.23 MeV (d) 3.64 MeV
9. The impact parameter for an alpha particle approaching a target nucleus is maximum when the scattering angle (θ) is :
- (a) 0° (b) 90°
(c) 180° (d) 45°
10. Two nuclei have their mass numbers in the ratio of 1 : 27. What is the ratio of their nuclear densities ?
- (a) 1 : 27 (b) 1 : 1
(c) 1 : 9 (d) 1 : 3

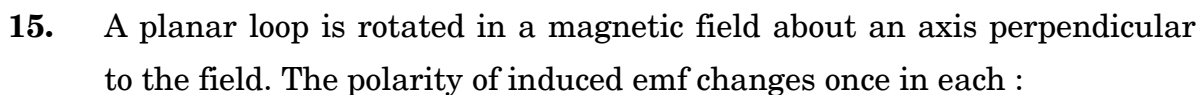
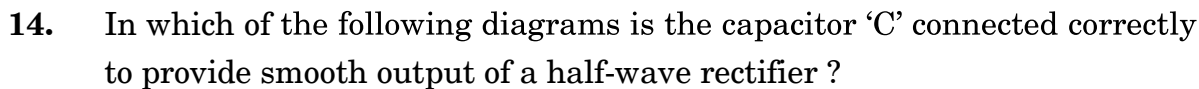


11. The given figure shows a capacitor C and a resistor R connected in series to an ac source. V_1 and V_2 are voltmeters and A is an ammeter.



Which of the following statements is correct ?

- (a) Current in the circuit lags in phase with voltage shown in V_2 .
 - (b) The voltage shown in V_1 is ahead in phase with voltage shown in V_2 .
 - (c) The current in the circuit and the voltage shown in V_1 are always in phase.
 - (d) The voltage shown in V_1 lags behind in phase with the voltage shown in V_2 .
12. A plane wave is incident on a concave mirror of radius of curvature R . The reflected wave is a spherical wave of radius :
- (a) $\frac{R}{4}$
 - (b) $\frac{R}{2}$
 - (c) R
 - (d) $2R$
13. Light of frequency $1.5 \nu_0$ is incident on a photosensitive material of threshold frequency ν_0 . If the frequency of the incident radiation is kept constant and intensity is increased, the photo current will :
- (a) increase
 - (b) decrease
 - (c) not change
 - (d) first decrease and then become zero



- Questions number **16** to **18** are Assertion (A) and Reason (R) type questions. Two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.

- 16.** *Assertion (A):* The temperature coefficient of resistance is positive for metals and negative for semi-conductors.

Reason (R): The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.



17. *Assertion (A)* : The magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area.

Reason (R): The magnetic field inside the solenoid is uniform.

18. *Assertion (A)* : Silicon is preferred over germanium for making semiconductor devices.

Reason (R) : The energy gap for germanium is more than the energy gap for silicon.

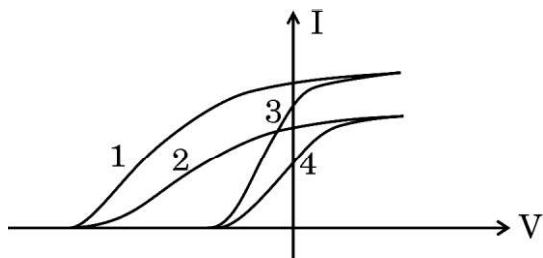
SECTION B

19. A bright point source is placed at the bottom of a tank containing a transparent liquid (refractive index μ) to a height H . Let r be the radius of the circular area of the liquid surface through which light can emerge out. Find the ratio (r/H). 2
20. (a) The electric field of an electromagnetic wave passing through vacuum is represented as $E_x = E_0 \sin (kz - \omega t)$. Identify the parameter which is related to the (i) wavelength, and (ii) the frequency of the wave in the above equation.
(b) Write two properties of a medium that determine the velocity of light in that medium. 2
21. A sphere S_1 of radius r_1 enclosing a charge Q_1 is surrounded by another concentric sphere S_2 of radius r_2 ($r_2 > r_1$). If there is a charge $-Q_2$ in the space between S_1 and S_2 , find the ratio of electric flux through S_1 and S_2 . 2
22. Light of wavelength $3500 \text{ \AA}$ is incident on two metals A and B. Which of them will yield photoelectrons, if their work functions are 4.2 eV and 1.9 eV respectively ? Make the necessary calculations to justify your answer. 2
23. (a) A uniform electric field E of 500 N/C is directed along $+x$ axis. O, B and A are three points in the field having x and y coordinates (in cm) $(0, 0)$, $(4, 0)$ and $(0, 3)$ respectively. Calculate the potential difference between the points (i) O and A, and (ii) O and B. 2

OR



- (b) Three point charges $1\ \mu\text{C}$, $-1\ \mu\text{C}$ and $2\ \mu\text{C}$ are kept at the vertices A, B and C respectively of an equilateral triangle of side 1 m. A_1 , B_1 and C_1 are the midpoints of the sides AB, BC and CA respectively. Calculate the net amount of work done in displacing the charge from A to A_1 , from B to B_1 and from C to C_1 . 2
24. (a) When is more power delivered to a light bulb — just after it is turned on and the glow of the filament is increasing or after the glow becomes steady ? Why ? 2
- OR**
- (b) A battery is connected first across the series combination and then across the parallel combination, of three resistances R, $2R$ and $3R$. In which of the three resistances will power dissipated be maximum in the two cases ? Justify your answer. 2
25. The figure shows the variation of photoelectric current (I) versus applied voltage (V) for two photosensitive materials and for two different intensities of the incident radiation. Identify the pairs of curves that correspond to different materials for same intensity of incident radiation. Justify your answers in brief. 2



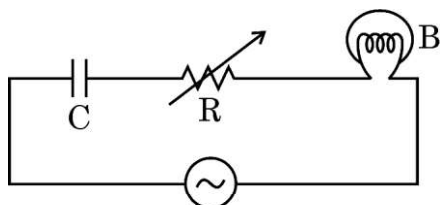
SECTION C

26. A horizontal straight metallic rod of length 4 m is held at some height above the surface of Earth, in east-west direction. If it is allowed to fall from rest, find the :
- emf induced in the rod 2 s after it starts falling,
 - polarity of the emf induced, and
 - the end of the rod which is at the higher potential.

The horizontal component of the Earth's magnetic field at the place is $0.3 \times 10^{-4}\ \text{Wb/m}^2$ and take $g = 10\ \text{m/s}^2$. 3



27. The figure shows a capacitor C , a variable resistor R and a bulb connected in series to the ac source of frequency (ν). The bulb glows with some brightness.



How will the glow of the bulb be affected, if the

- (a) separation between the plates of the capacitor is doubled, keeping resistance R and frequency (ν) the same ?
 - (b) resistance R is decreased keeping the value of capacitance C and frequency (ν) same ?
 - (c) frequency of ac source is decreased for the same value of C and R ?
- Justify your answer in each case.

3

28. (a) (i) An electron in a hydrogen atom jumps from second excited state to the first excited state. Name the spectral series in the spectrum of hydrogen atom to which the emitted radiation belongs.
- (ii) Find the ratio of the wavelengths of the “most energetic spectral” lines in the Balmer series to that in Paschen series of Hydrogen spectrum.

3

OR

- (b) (i) An α -particle having kinetic energy K approaches a nucleus of atomic number Z . It gets close to the nucleus and then approaches a distance (d) and reverses its direction. Obtain an expression for the distance of closest approach (d) in terms of kinetic energy of the α -particle.
- (ii) A proton and an alpha particle approach a target nucleus in head-on position, with equal velocities. Find the ratio of their distances of closest approach to the target nucleus.

3



29. (a) (i) In which case is diffraction effect more dominant — slit formed by 2 blades or slit formed by two fingers ?
- (ii) Yellow light ($\lambda = 6000 \text{ \AA}$) illuminates a single slit of width $1 \times 10^{-4} \text{ m}$. Calculate (i) the distance between two dark lines on either side of central maximum, in the diffraction pattern observed on a screen kept 1.5 m away from the slit, and (ii) the angular spread of the first minimum. 3

OR

- (b) (i) What will be the colour of the central bright fringe in Young's double slit experiment if the monochromatic source is replaced by a source of white light ? Give reason for your answer.
- (ii) In Young's double slit experiment, the slits are separated by 0.3 mm and the screen is placed 1.5 m away from the slits. The distance between the central bright fringe and the sixth bright fringe is found to be 1.8 cm. Find the wavelength of light used in the experiment. 3

30. (a) It is not advisable to use a galvanometer as such to measure current directly. Why ?
- (b) Why should the value of resistance connected in parallel to a galvanometer be low ?
- (c) Is the reading shown by an ammeter in a circuit less than or more than the actual value of current flowing in the circuit ? Why ? 3

SECTION D

31. (a) Draw a ray diagram for formation of a real and diminished image of an object kept in front of a concave mirror. Hence derive the mirror equation.
- (b) A concave mirror of focal length 10 cm produces a real image which is 3 times the size of the object. Find the distance of the object from the mirror. 5

OR

- (a) Two converging lenses of focal lengths f_1 and f_2 are placed coaxially in contact. Derive the expression for the focal length of the combination using the ray diagram for the formation of the image of an object by the combination.

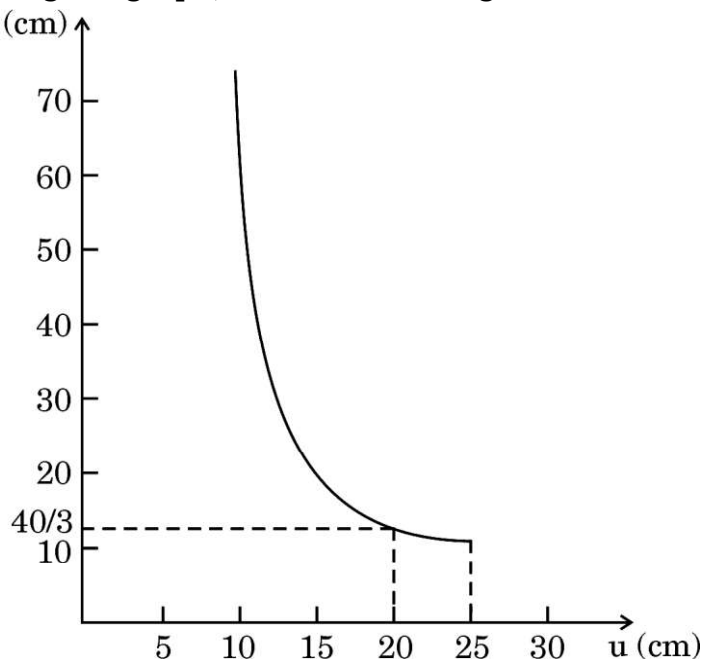


(b) The figure shows the variation of image distance v with object distance u for a lens.

(i) What is the nature of the lens ?

(ii) Using the graph, find the focal length of the lens.

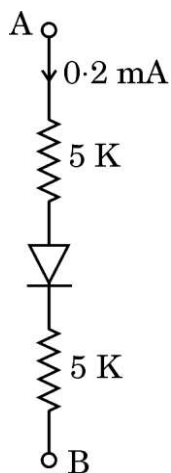
5



32. (a) (i) Draw the circuit diagram used to study I – V characteristics of a p-n junction diode in conducting mode. Mark on the graph the threshold voltage of the diode. Explain the significance of this voltage.

(ii) In the circuit shown in the figure, the forward voltage drop across the diode is 0.3 V. Find the voltage difference between A and B.

5



OR

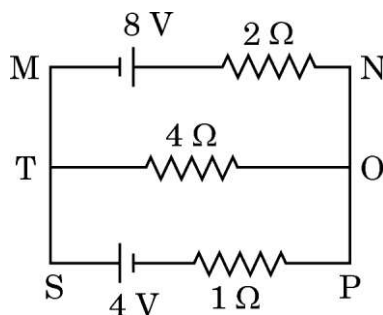


- (b) (i) Briefly describe the classification of solids into metals, insulators and semi-conductors on the basis of energy level diagrams.
- (ii) In a silicon diode, the current increases from 10 mA to 20 mA when the voltage changes from 0.6 V to 0.7 V. Calculate the dynamic resistance of the diode. 5

33. (a) (i) Derive the relation between the current and the drift velocity of free electrons in a conductor. Briefly explain the variation of resistance of a conductor with rise in temperature.
- (ii) An ammeter, together with an unknown resistance in series is connected across two identical batteries, each of emf 1.5 V, connected (i) in series, and (ii) in parallel. If the current recorded in the two cases be $\left(\frac{1}{2}\right)$ A and $\left(\frac{1}{3}\right)$ A respectively, calculate the internal resistance of each battery. 5

OR

- (b) (i) State Kirchhoff's rules. Use them to obtain the condition of balance for a Wheatstone Bridge.
- (ii) Use Kirchhoff's rule to determine the currents flowing through the branches MN, TO and SP in the circuit shown in the figure. 5





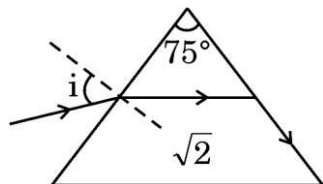
SECTION E

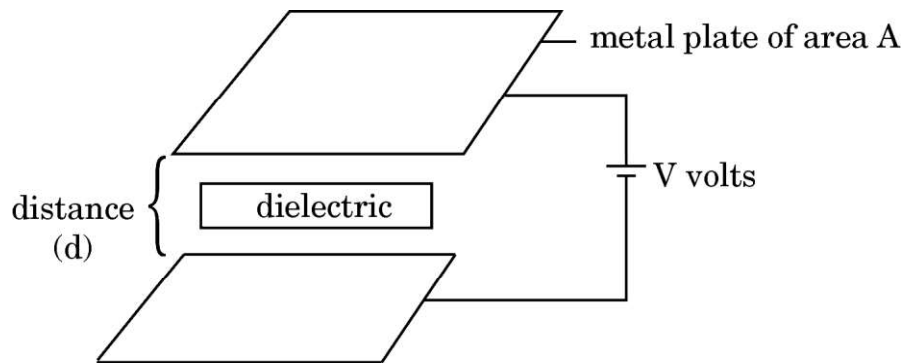
34. A prism is a solid transparent medium bounded by three rectangular faces with a triangular base and a top. A ray of light incident at angle i on one face of a prism suffers two refractions on passing through a prism. Hence it deviates through a certain angle δ from its original path. The angle of deviation becomes minimum ($\delta = \delta_m$) for a certain value of angle i . In such a condition, the refracted ray inside the prism becomes parallel to its base. An expression for refractive index μ of the material of the prism can be obtained in terms of angle A and angle δ_m .

- (a) Show in a figure the variation of angle δ with angle of incidence i . 1
- (b) Show that for a prism of small angle A , the refractive index μ of its material can be written as $\mu = 1 + \frac{\delta_m}{A}$. 1
- (c) A ray of light passes through an equilateral prism such that both the angle of incidence and the angle of emergence are equal to the angle of prism A . Find the refractive index of the material of the prism, in terms of A . 2

OR

- (c) A ray of light passes through a prism of angle 75° , as shown in the figure. The refractive index of the material of the prism, with respect to its surrounding is $\sqrt{2}$. Find the angle of incidence i . 2





A parallel plate capacitor is an arrangement of two identical metal plates kept parallel, a small distance apart. The capacitance of a capacitor depends on the size and separation of the two plates and also on the dielectric constant of the medium between the plates. Like resistors, capacitors can also be arranged in series or parallel or a combination of both. By virtue of electric field between the plates, charged capacitors store energy.

- The capacitance of a parallel plate capacitor increases from $10\text{ }\mu\text{F}$ to $80\text{ }\mu\text{F}$ on introducing a dielectric medium between the plates. Find the dielectric constant of the medium.
- n capacitors, each of capacitance C , are connected in series. Find the equivalent capacitance of the combination.
- A capacitor is charged to a potential (V) by connecting it to a battery. After some time, the battery is disconnected and a dielectric is introduced between the plates. How will the potential difference between the plates, and the energy stored in it be affected? Justify your answer.

OR

- (c) Find the equivalent capacitance between points A and B, if capacitance of each capacitor is C.

